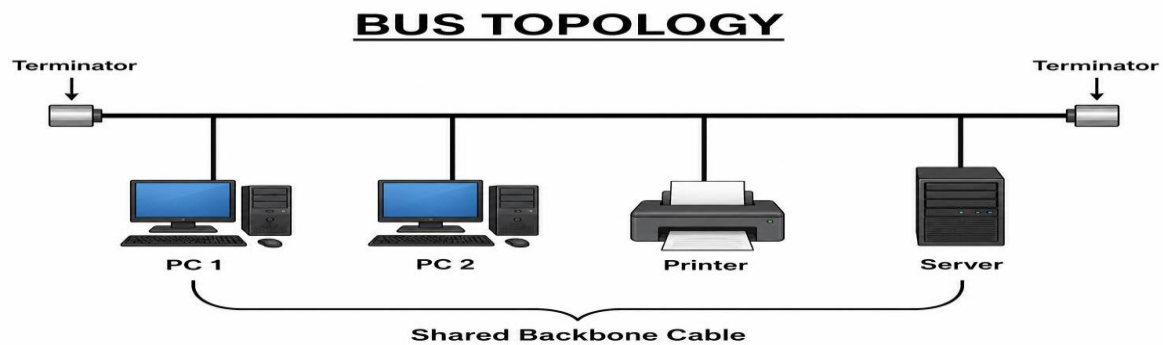


1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.



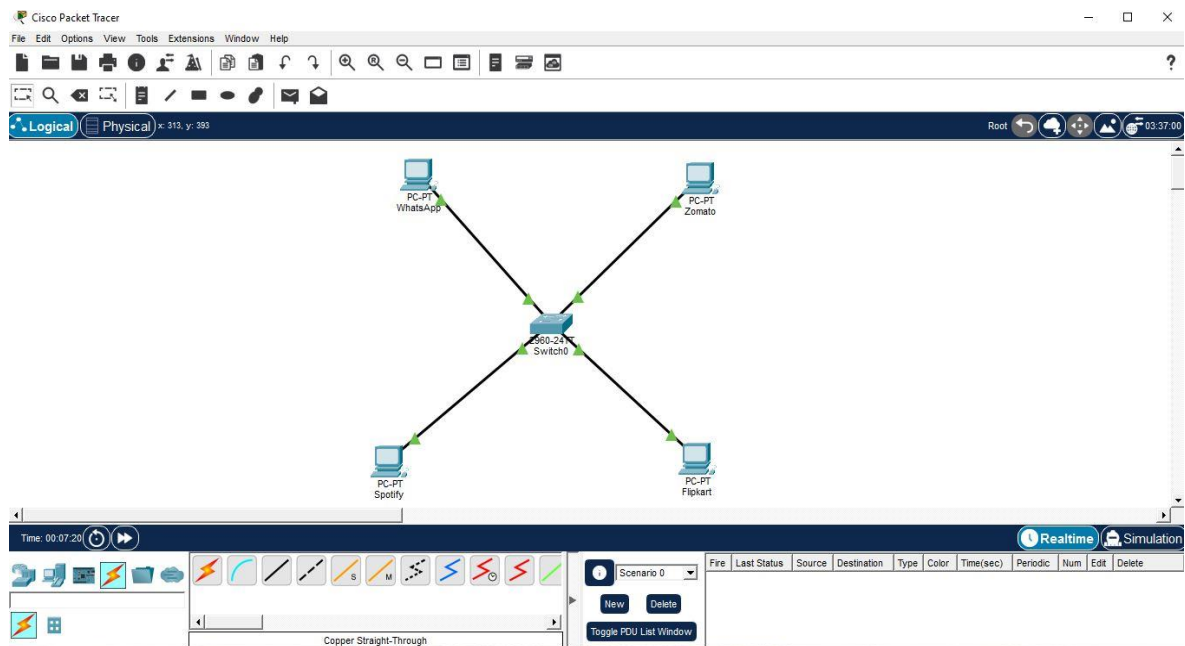
- **Key Components of Bus Topology**

- Backbone Cable – The main communication cable that connects all devices.
- Terminator – Placed at both ends of the backbone cable to prevent signal reflection.
- PC1 and PC2 – Computers connected to the shared cable.
- Printer – A shared network printer connected to the bus.
- Server – Provides shared files, applications, or network services to the connected devices.

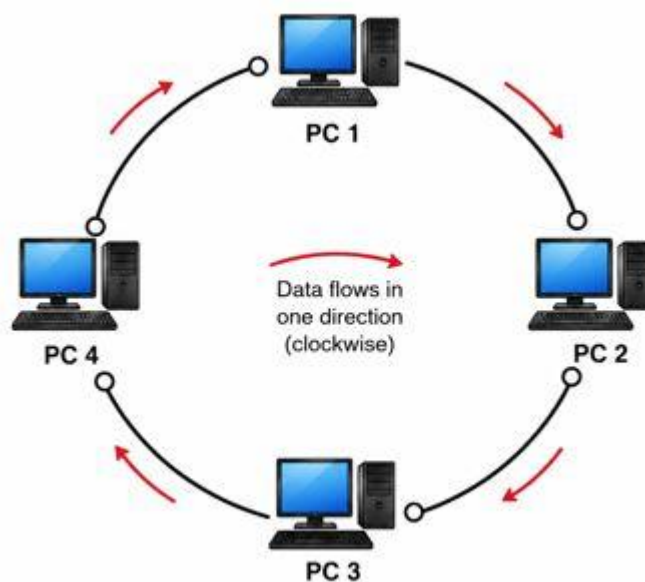
- **Real-world example**

- A small school computer lab with a limited number of computers can use a bus topology. All computers are connected to a single backbone cable, allowing students to access shared files and printers. Although it is inexpensive and easy to set up, bus topology is rarely used today because if the main cable fails, the entire network stops working.

2. In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, and Flipkart) and take a screenshot of your topology.



3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.



- In the ring topology, each device is connected to exactly two neighbouring devices, forming a closed loop. Data travels around the ring in the direction of the arrows, passing through each device until it reaches the intended destination. The organized path helps reduce data collisions because only one device typically transmits data at a time.

4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Feature	Mesh Topology	Star Topology
Advantages	1. Highly reliable because multiple paths exists between devices. 2. Failure of one connection usually does not affect the rest of the network.	1. Easy to install, manage and expand. 2. Failure of one device or cable does not affect other connected devices.
Disadvantages	1. Expensive due to the large number of cables and connections required. 2. Difficult to install and maintain because of its complexity.	1. If the central switch or hub fails, the entire network stops working. 2. Requires more cable than a bus topology, increasing installation cost.